



Algorithm and Data Structures

26.1 Batch - Faculty of Computing

1. Arrange the following numbers in ascending order using **Bubble Sort**. Show the array after each pass.
8, 3, 6, 2, 5
2. Arrange the following numbers in ascending order using **Selection Sort**. Show the array after each pass.
7, 4, 9, 1, 6
3. What is the main difference between **Bubble Sort** and **Selection Sort**?
4. In Bubble Sort, what happens when two adjacent elements are in the wrong order?
5. In Selection Sort, how is the correct element selected during each pass?
6. Apply **Bubble Sort** to the following array for **two passes only**.
10, 7, 3, 8, 2
7. Apply **Selection Sort** to the following array for **three passes only**.
12, 5, 9, 3, 7, 1
8. Write an algorithm or pseudocode for sorting an array in ascending order using **Bubble Sort**.
9. Write an algorithm or pseudocode for sorting an array in descending order using **Selection Sort**.
10. Consider the array:
6, 2, 8, 4, 1
Sort it using both Bubble Sort and Selection Sort. Compare the number of swaps performed by the two algorithms